American Computer Science League

2024-2025 ● Contest 2: Short Problems Solutions ● Junior Division

1. Prefix/Infix/Postfix Notation

3 + 4 * (8 − 4 / 2)
= 3 + 4 * (8 - (/42))
= 3 + 4 * (- 8 / 4 2)
= 3 + (* 4 - 8 / 4 2)
= + 3 * 4 - 8 / 4 2

B. + 3 * 4 − 8 / 4 2

2. Prefix/Infix/Postfix Notation

9 8 6 2 − * 2 / + 6 3 / −
= 9 8 (6 2 −) * 2 / + (6 3 /) −
= 9 (8 4 *) 2 / + 2 −
= 9 (32 2 /) + 2 −
= (9 16 +) 2 −
= 25 2 −
= 23

D. 23

3. Bit-String Flicking

01011 OR NOT 01101
= 01011 OR (NOT 01101)
= 01011 OR 10010
= 11011

A. 11011

4. Bit-String Flicking

(LSHIFT-2 01101) OR (RCIRC-3 10110) AND (RSHIFT-1 11011)
= 10100 OR (11010 AND 01101)
= 10100 OR 01000
= 11100

C. 11100

5. What Does This Program Do? (Loops)

This is Euclid's algorithm that finds the GCF in 2 iterations of the loop. Trace the program as follows:

```
m = 50
n = 40
while n != 0
    r = m % n
    m = n
    n = r
end while
output m
```

m	n	r
50	40	10
40	10	0
10	0	GCF=10

GCF(50,40) = GCF(40,10) = 10

D. 10

6. What Does This Program Do? (Loops)

This program finds the sum of the numbers generated by the 2 loops:

```
s = 0
for y = 2 to 8 step 2
    s = s + y / 2
next y
for z = 3 to 1 step -1
    s = s + z ^ 2
next z
output s
```

y	s	z	s
2	1		
4	3		
6	6		
8	10		
		3	19
		2	23
		1	24

A. 24